



GLOBAL ECONOMICS
GRADE: 11TH
TERM 3 — LEARNING EVIDENCE 4
WEB IMPLEMENTATION AND REPORT OF
HYPOTHESIS TESTING (C8–C10)
TEACHER’S NAME: Nicolás López Cuéllar

THIRD
TERM
2025–2026

Learning objective: Build a functional webpage that extends the C1–C7 hypothesis-testing process into an interactive digital tool, including manual parameter control, visual p-value interpretation on a normal curve, automatic analysis from uploaded two-group CSV data, and a short evidence-based report using real-world sources.

Assessed criteria:

C8: Creates an interactive webpage that represents the p-value of a hypothesis test on a normal curve and allows the user to modify the test parameters.
C9: Creates a webpage that imports two-group CSV data and automatically performs a hypothesis test using the uploaded data.
C10: Uses hypothesis testing and real-world sources to write an evidence-based report with statistical results and contextual interpretation.

Background from C1–C7

In C1–C7 you already practiced the full statistical reasoning cycle: writing H_0 and H_1 , selecting the tail direction, calculating test statistics, making decisions with α , interpreting errors, and determining minimum sample sizes. In this activity, you will transform that reasoning into a working webpage and use the results to write a short report.

Task overview

Create one webpage (HTML + CSS + JavaScript) that performs hypothesis testing in two connected modes, and use it to support a short written report:

1. **Manual mode (C8):** users set parameters and visualize the p-value on an interactive normal curve.
2. **CSV mode (C9):** users upload two-group data and the webpage computes the hypothesis test automatically.
3. **Report work (C10):** students write a short report using the webpage results and three real-world sources.

The page must be functional, not only visual. Do not submit worked solutions, completed answer sheets, or screenshots as a replacement for functionality.

GitHub and deployment tutorials

Use these tutorials if you need help preparing and publishing your webpage:

1. [Tutorial to download Git and push a project into GitHub.](#)
2. [Tutorial to host a website using GitHub Pages.](#)

Required AI conversation links

For the work to be valid, the deployed webpage must include a visible section titled **AI development record** or **AI conversations used**. This section must contain hyperlinks to the AI conversation or conversations used to help create the webpage. These links must be accessible directly from the deployed webpage, not only from the written report.

If the deployed webpage does not include these AI conversation links, the C8–C9 implementation will be considered incomplete.

Mandatory requirements for C8

Your C8 implementation must include all of the following:

1. Input controls using both **sliders** and **input boxes**.
2. Test-direction selector: left-tailed, right-tailed, two-tailed.
3. Significance level α .
4. Two input modes for calculating the test statistic z :
 - **Proportion setting mode:** the webpage must calculate z using the benchmark proportion, sample proportion, and sample size.
 - **Mean setting mode:** the webpage must calculate z using the benchmark mean, sample mean, standard deviation or standard error, and sample size.

Each parameter in both modes must be editable using both sliders and input boxes.

5. Null value, hypothesis or benchmark.
6. Sample statistic.
7. A hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.
8. A visible **AI development record** section in the deployed webpage with hyperlinks to the AI conversation(s) used.

The output panel/graph must show:

1. The normal curve.
2. The observed test statistic.
3. The shaded p-value region.
4. The numerical p-value.
5. The critical value or critical region.
6. The statistical decision: reject H_0 or fail to reject H_0 .
7. A short contextual conclusion consistent with the decision.

Mandatory requirements for C9

Your C9 implementation must include all of the following:

1. CSV upload control.
2. Reading data from two groups.
3. Detection of group columns, or user selection of group columns.
4. Computation of summary statistics for each group.
5. Selection of benchmark group and test group.
6. Test-direction selector: left-tailed, right-tailed, two-tailed.

7. Automatic execution of the hypothesis test.
8. Display of test statistic, p-value, decision, and contextual conclusion.
9. Display of the p-value on the same type of normal-curve graph used in C8.
10. A hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.
11. A visible **AI development record** section in the deployed webpage with hyperlinks to the AI conversation(s) used.

Required CSV example files for C9

CSV datasets

Use the following test CSV databases to verify the C9 CSV mode:

- [Test database 1.](#)
- [Test database 2.](#)
- [Test database 3.](#)

Expected structure:

- Two-group CSV data using a group variable and a numerical variable, or a binary success variable.

These datasets are provided to test the C9 CSV mode. Your webpage must be able to upload each file, identify or allow selection of the two groups, compute the required summary statistics, perform the hypothesis test, show the p-value on the normal curve, and display a statistical decision. You may use additional datasets, but these three files must work correctly in your webpage.

Include screenshots or equivalent evidence in the C10 report showing that each provided CSV file was tested in your C9 webpage.

Mandatory requirements for C10

Your C10 work must be a short report based on hypothesis testing and real-world context. The report must follow APA style and include all of the following:

1. A title.
2. A real-world context connected to the hypothesis test.
3. At least **three real-world sources**.
4. A clear research question.
5. The null and alternative hypotheses.
6. The test direction and significance level α .
7. The test statistic, p-value, and statistical decision.
8. Hyperlink to the deployed webpage (GitHub Pages or another accessible public/shared link).
9. Screenshots showing the provided CSV files being used.
10. A short explanation of how the CSV data was uploaded.

11. A short section referencing the AI conversation links included in the deployed webpage.
12. A contextual conclusion consistent with the decision.
13. A short discussion of what the result means, and the errors that can be incurred in the real-world context.
14. APA-style in-text citations when using information from the sources.
15. A references section in APA format.

APA format requirements for C10

The report must use APA style in the following way:

1. Use clear academic headings, such as title, introduction, methodology, results, discussion, conclusion, and references.
2. Use author–year in-text citations when making contextual claims. Example: (Author, year).
3. Include a references section titled **References**.
4. List the references in alphabetical order by the first author’s last name or by the institutional author.
5. Each reference must include the author or institution, year, title, source name, and URL or DOI when available.
6. Use APA 7th edition style for web sources. Example format: Author or Institution. (Year). *Title of page or report*. Website or Publisher. URL
7. Do not include sources in the references section if they are not cited inside the report.

Real-world source requirements for C10

The three real-world sources must satisfy the following:

1. They must be connected to the context of the report.
2. They must be reliable sources, such as official statistics sites, academic publications, institutional reports, or reputable data portals.
3. They must be cited inside the report when making contextual claims.
4. They must appear in the references section.

Deliverables and evaluation checklist by criterion

Submit and verify the following items:

1. **C8: Manual webpage mode**
 - Submit `index.html` or an equivalent main HTML file.
 - Submit linked stylesheet file(s) and JavaScript file(s).
 - Submit the hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.
 - The deployed webpage must include visible hyperlinks to the AI conversation(s) used.
 - The page must include sliders and input boxes for manual parameter control.

- The graph, p-value, and decision must update when parameters change.
- The output must show the normal curve, observed test statistic, shaded p-value region, numerical p-value, critical value or critical region, decision, and types of errors.

2. C9: CSV webpage mode

- The webpage must be validated with the three provided CSV files (dataset 1, dataset 2, dataset 3).
- Submit the hyperlink to the webpage deployed through GitHub Pages or another accessible public/shared link.
- The deployed webpage must include visible hyperlinks to the AI conversation(s) used.
- Submit a short README explaining how to open the webpage and test C9 mode with the three provided CSV files.
- The CSV files must be read correctly.
- The test must be computed correctly from uploaded data.
- The three provided CSV files must work in the webpage. Additional datasets are allowed.
- The displayed test statistic, p-value, decision, graph, and contextual conclusion must be consistent.

3. C10: Report

- Submit the C10 report as a PDF or source document.
- Include the hyperlink to the deployed webpage.
- Submit or clearly reference the three real-world sources used in the report.
- The report must include a research question, hypotheses, test direction, significance level, test statistic, p-value, decision, and contextual conclusion.
- The report must include screenshots showing the provided CSV files being used and interpreted.
- The report must include a short explanation of how the CSV data was uploaded.
- The report must include a short section referencing the AI conversation links included in the deployed webpage.
- The report must use the three sources to support the real-world context.
- The report must use APA-style in-text citations and APA-style references.
- The conclusion must match the statistical decision.
- The report must include a references section.